# Which rating does Maia-2 need for this player?

Fitting the chess.com to Lichess offset on the player's own games

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**Abstract**

The coaching app converts the player's chess.com rating to Maia-2's Lichess scale by adding 250 points, a number its own code calls a default, not a measurement. We measured it on the 648 rapid games of the player's archive: at every offset from 0 to 500, how likely Maia-2 finds the 33,522 moves both players made. By a rule fixed before any number was computed, the fitting half of the games chose 250, the app's own number, so it stays; the held-out half chose it too. Beside the verdict, a single offset hides a split: Maia-2 predicts the player's own moves best at 175 and his opponents' at 425, with intervals that do not overlap. Whether separate offsets beat one is a question for its own rule.

## 1 The question

Maia-2 predicts the move a human plays from the position and two ratings, the mover's and the opponent's. It learned from Lichess games, so its ratings are Lichess ratings, in 11 bands: one below 1100, one for every 100 points up to 2000, and one from 2000 up. The coaching app knows the player's chess.com rating and adds 250 points to it before it asks Maia-2 anything; its own code calls that number a default, not a measurement. Every Maia number the app gives rests on it: the odds that a mistake gets punished, how findable a move is at the player's level, and the Maia bot's model of the player and of the opponent it plays as.

The player's games can measure it. The offset Maia-2 needs is the one under which it finds the moves actually played most likely, with both players at their own chess.com ratings converted by that offset.

## 2 The moves

The games are the 648 in the app's archive, every one from the standard start with both ratings and a clock after every ply. All are rapid: 647 at ten minutes a side and one at a slower control. A move counts from ply 11 on when both players had at least 30 seconds left at the position, the protocol of the Maia papers that the lab's Maia-2 benchmark also follows. That leaves 33,522 moves: 16,766 by the player and 16,756 by his opponents. The movers are rated from 513 to 1228 on chess.com (median 1015, which the app's offset puts in Maia's 1200 band); the rating is the one chess.com recorded after each game.

## 3 The instrument and the rule

At an offset $o$, a move by a player rated $r$ against one rated $r'$ scores the logarithm of Maia-2's probability of the move played, with the mover's band the one $r + o$ falls in and the opponent's the one $r' + o$ falls in, exactly as the app computes them (`defaultMaiaElo`), through the app's own Maia-2 in Node (`humanPolicy` with both bands). The offsets run over the range the app allows, 0 to 500, in steps of 25 (21 offsets). The bands are 100 points wide, so a step changes only the players whose converted rating crosses a band edge, and at the app's offset everyone rated under 850 on chess.com shares the lowest band.

The rule was committed before any number was computed. The games, ordered by when they ended, alternate between a fitting half (324 games, 17,055 moves) and a held-out half (324 games, 16,467 moves). The fitted offset is the one with the highest mean log-likelihood per move on the fitting half. It replaces 250 only if, on the held-out half, its mean log-likelihood per move beats 250's with the 95% interval of the difference, over 1,000 bootstrap draws of the held-out games, entirely above zero.

## 4 Results

The fitting half's best offset is 250, the app's own, so 250 stays. The held-out half's best is 250 as well, and so is the best over all the games; over 1,000 bootstrap draws of the games, the best offset falls between 250 and 275 (Table 1, Figure 1).

Near the top the offset matters little. At 250 the moves played average a log-likelihood of −1.396 per move; 0 loses 0.0215 per move and 500 loses 0.0142. As a geometric mean, Maia-2 gives the moves played 2.2% more probability at 250 than at 0, and 1.4% more than at 500. Its first choice is the move played 53.4% to 53.9% of the time across the whole range: the offset changes how sure Maia-2 is far more than which move it ranks first.

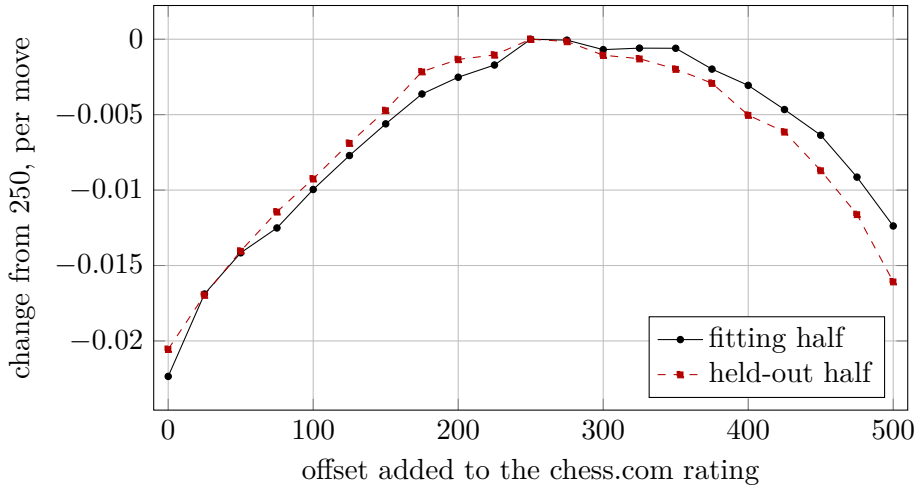



Figure 1: The mean log-likelihood per move of the moves played at each offset, as a change from its value at the app's 250. Both halves peak at 250.

| offset | fitting half | held-out half | all games | first choice |
|---|---|---|---|---|
| 0 | −1.4182 | −1.4164 | −1.4173 | 53.4% |
| 25 | −1.4128 | −1.4128 | −1.4128 | 53.5% |
| 50 | −1.4100 | −1.4099 | −1.4100 | 53.6% |
| 75 | −1.4084 | −1.4073 | −1.4079 | 53.7% |
| 100 | −1.4058 | −1.4051 | −1.4055 | 53.7% |
| 125 | −1.4036 | −1.4028 | −1.4032 | 53.7% |
| 150 | −1.4015 | −1.4006 | −1.4011 | 53.7% |
| 175 | −1.3995 | −1.3980 | −1.3988 | 53.8% |
| 200 | −1.3984 | −1.3972 | −1.3978 | 53.8% |
| 225 | −1.3976 | −1.3969 | −1.3973 | 53.9% |
| 250 (app, fitted) | −1.3959 | −1.3959 | −1.3959 | 53.9% |
| 275 | −1.3959 | −1.3960 | −1.3960 | 53.9% |
| 300 | −1.3966 | −1.3969 | −1.3967 | 53.9% |
| 325 | −1.3965 | −1.3972 | −1.3968 | 53.9% |
| 350 | −1.3965 | −1.3979 | −1.3972 | 53.9% |
| 375 | −1.3979 | −1.3988 | −1.3983 | 53.8% |
| 400 | −1.3989 | −1.4009 | −1.3999 | 53.8% |
| 425 | −1.4005 | −1.4020 | −1.4013 | 53.7% |
| 450 | −1.4022 | −1.4046 | −1.4034 | 53.7% |
| 475 | −1.4050 | −1.4075 | −1.4062 | 53.6% |
| 500 | −1.4083 | −1.4119 | −1.4101 | 53.6% |

Table 1: The mean log-likelihood per move at each offset, on each half and over all the games, and the share of all the moves that are Maia-2's first choice.

## 5 Beside the verdict

Named before the first run, these looks decide nothing, but they show what one offset hides (Table 2, Figure 2). Maia-2 predicts the player's own moves best at 175 (150 to 200 over the draws) and his opponents' best at 425 (350 to 450); the two intervals do not overlap. By the movers' rating, the weakest third (513 to 934) fits best at 400, and the two stronger thirds at 250, like the moves as a whole. The gains a part's own best offset would bring over 250 are small: 0.0018 per move for the player's moves, 0.0064 for his opponents' and 0.0042 for the weakest third.

One offset moves both players at once, so it cannot follow the split: every offset that fits his opponents better than 250 fits him worse, and every one that fits him better fits them worse. 250 is where the two balance. Whether an offset for the player and another for his opponents would beat one on held-out games is a question for its own project, with its rule written first: these looks chose where to look, so they cannot also be the test. A split would also move both directions of the punish odds, which the punish-odds project found calibrated at 250, so it would have to keep them so.

Figure 3 shows what the bands do in one position: the player's move whose probability moves most between 0 and 500. The stronger the bands the offset gives the two players, the surer Maia-2 is of the move he chose.

| moves | count | best offset | 95% interval over draws |
|---|---|---|---|
| all | 33,522 | 250 | 250 to 275 |
| the player's | 16,766 | 175 | 150 to 200 |
| his opponents' | 16,756 | 425 | 350 to 450 |
| movers rated 513 to 934 | 10,783 | 400 | 325 to 425 |
| movers rated 935 to 1075 | 11,292 | 250 | 200 to 275 |
| movers rated 1076 to 1228 | 11,447 | 250 | 225 to 275 |

Table 2: The best offset for each part of the moves, and its 95% interval over 1,000 bootstrap draws of the games. The thirds split the movers' chess.com ratings.

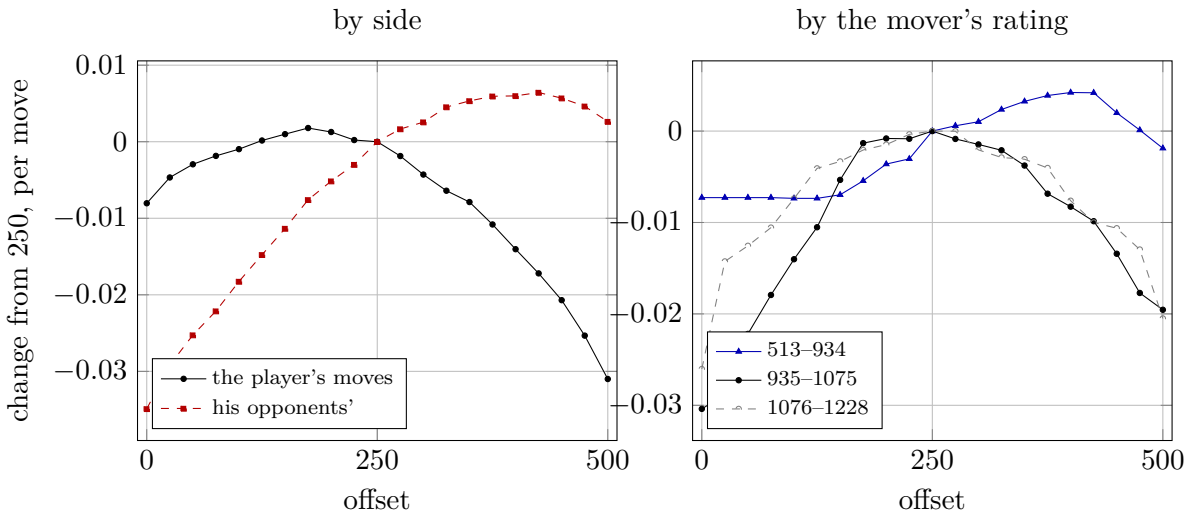
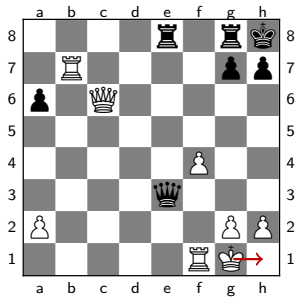



Figure 2: The same curve for each part of the moves, as a change from each part's value at 250.



**Played:** 29. ♔h1 (red), by the player at 1061 against 1082.

| offset | his band | opponent's | played | Maia-2's first choice |
|---|---|---|---|---|
| 0 | <1100 | <1100 | 40.6% | 29. ♖f2, 59.4% |
| 250 | 1300 | 1300 | 83.1% | 29. ♔h1, 83.1% |
| 500 | 1500 | 1500 | 95.2% | 29. ♔h1, 95.2% |

**UCI:** played `g1h1`; first choices `f1f2`, `g1h1`, `g1h1`
**FEN:** `4r1rk/1R4pp/p1Q5/8/5P2/4q3/P5PP/5RK1 w - - 0 29`

Figure 3: The player's move at the bands three offsets give the two players: Maia-2's probability of the move played and its first choice.

## 6 What it means for the app

Nothing to port. The app's 250 is the offset this player's games choose, so the punish odds, how findable a move is and the Maia bot stand as they are. The app lets the player change the offset; any offset from 250 to 275, the range the best one takes over the draws, fits these games about as well. The split between the player and his opponents is the next question for this part of the app.

## 7 Limitations

- One player's games and his opponents', rated 513 to 1228 on chess.com. The offset that fits them says nothing about players far from that range.
- The rating is chess.com's after each game, a few points from the one the game was played at.
- Maia-2's lowest band holds everyone under 1100 on its scale, so it cannot tell the weakest players here apart from each other at any offset that leaves them there.
- The likelihood measures how well Maia-2 predicts the moves, not why. An offset that fits best also absorbs whatever Maia-2 gets wrong about this pool of players, which is what the app needs from it, but it is not a statement about the two sites' rating pools.